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A reader's guide to the empirical and pricing code

These guides describe the source present on November 1, 2026, including current working-tree changes. Read them in pipeline order:

1. Data construction and descriptive statistics.
2. Demand estimation and inference.
3. Dynamic pricing and comparative statics.

Each chapter connects mathematical definitions to functions, intermediate tables, algorithms, outputs, examples, and interpretation. The guides document the code; they do not certify the private server data or the model assumptions.

PDF editions

Read the complete collected edition, or the separate data guide, estimation guide, and supply guide. The overview PDF is also available. PDFs include a generated table of contents, bookmarks, typeset equations, wrapped code examples, and page numbers.

Rebuild all five PDFs from the repository's `Code` directory with `python3 final_draft/build_code_guides.py`. This uses local Pandoc and XeLaTeX and reads no private data. The Markdown files are the editable source.

How the components connect

```
Panel TSVs -> conversion/concatenation/merge -> attach week_end --+
                                                                   |
Retail TSVs -> conversion/concatenation/merge ------------------+
                                                                   v
                                                master_dat.py: matched purchases
                                                                   |
                                                summ_stats.py: prepared data
                                                                   |
                                                est.py: likelihood and inference

Fixed type + calibration -> supply backward induction -> derivatives/figures
```

The arrow from estimates to supply is deliberately absent: the current supply entry point does not load demand estimates. A researcher must explicitly define that bridge. Demand has three inside flavor groups and a cumulative state index; supply has one inside product, an outside choice, two periods, and signed last-choice states. The two gamma parameters therefore do not share a mechanical unit conversion.

Three meanings of a sample

The full panel consists of recorded purchases in `merged_panel.parquet`; it is not a roster of every enrolled household including households with no purchases. The matched yogurt sample contains purchases that survive retail matching, positive-price filtering, and the two-purchase-week restriction. The final estimation sample further requires compatible exposure-store menus and usable instruments, then reapplies the two-usable-week restriction. Its household count can be smaller than the descriptive sample's count.

Reading existing documentation

Use the Python source to settle runtime defaults. The estimation README contains older statements followed by amendments: current bootstrap replications default to **49**, simulation draws to **128**, and main starts to **3**. For supply, `MODEL_CONVENTIONS.md` and `comparative_statics/canonical.py` describe the current symmetric-switching model. Several other supply notes and retained output files describe superseded models. The chapters below identify those boundaries.